

AI CENTER
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Coalitional Games for Explainable Machine Learning 2

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Explainable Models Based on Coalitional Games

Supervised ML Models

Terminology

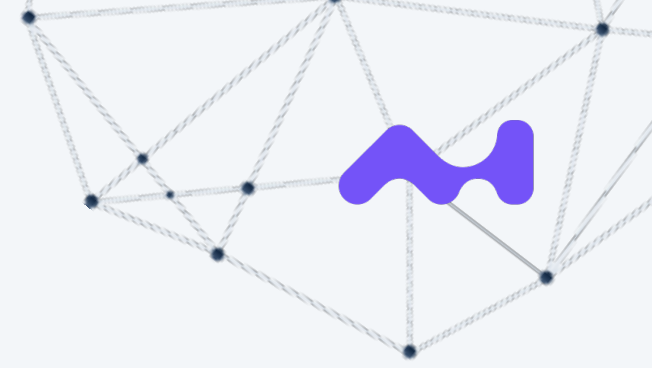
- Features $X_1, \dots, X_n$
- Target variable Y
- Model $f: \mathcal{X}_1 \times \dots \times \mathcal{X}_n \rightarrow \mathcal{Y}$

The goal of ML

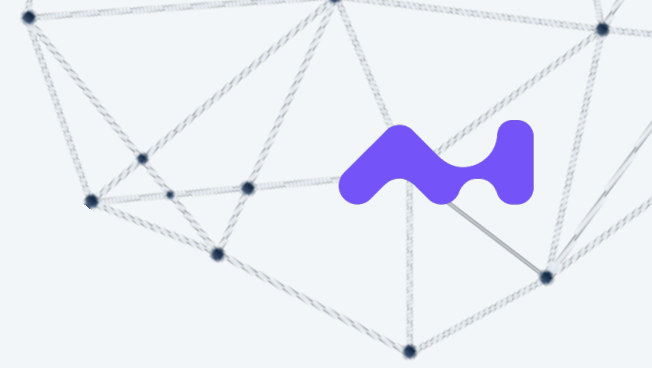
Given data $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_d, y_d)$,
find the best model f for the data.

In the following we assume:

We have already found the model f .



Characteristics of Explainable Methods



Model

- agnostic
- specific

Explaining

- a single prediction (local)
- the model (global)

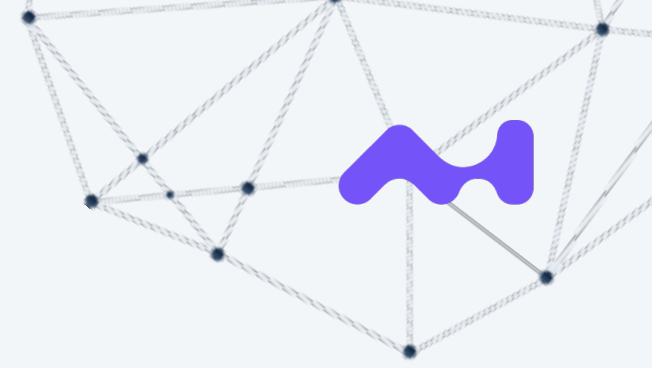
Explainable method

- game-theoretic
- information-theoretic
- counterfactual-based
- surrogate-based
- gradient-based

In the following we focus on:

Agnostic, local, and game-theoretic methods.

Explaining Supervised ML Models, Formally



For each data instance $\mathbf{x}$, find a vector $\varphi \in \mathbb{R}^n$ attributing an *influence* φ_i to each feature X_i on prediction $f(\mathbf{x})$.

Minimal requirements

- Irrelevant features have zero influence (*null player property*)
- Equally important features have the same influence (*symmetry*)

Coalitional Games for Model Explanations



- Features $X_1, \dots, X_n$ correspond to **players** in $N = \{1, \dots, n\}$
- Our goal is to evaluate the effect of **coalition** of features $A \subseteq N$ on $f(\mathbf{x})$
- We define a **coalitional game** $v_{\mathbf{x}}$ depending on f , the data distribution and $\mathbf{x}$

The number $v_{\mathbf{x}}(A)$ is the expected prediction if only the features in A are known.

Most methods fix $\mathbf{x}_A = (x_i)_{i \in A}$ and average out the effect of $\mathbf{X}_{\bar{A}} = (X_i)_{i \in \bar{A}}$.

IME (Interactions Method for Explanation)

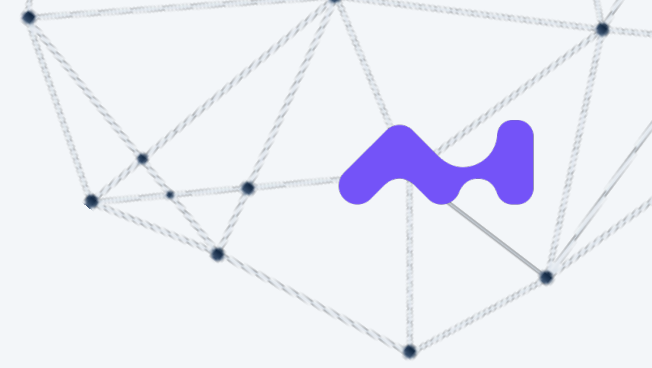


E. Strumbelj and I. Kononenko. Explaining prediction models and individual predictions with feature contributions. *Knowledge and Information Systems*, 41(3):647–665, 2014.

$$v_{\mathbf{x}}(A) = \mathbb{E}_{p_{\bar{A}}} [f(\mathbf{x}_A, \mathbf{X}_{\bar{A}})] - \mathbb{E}_p [f(\mathbf{X})]$$

- The features in $\bar{A}$ can be marginalized out with the uniform distribution

SHAP (SHapley Additive exPlanations)



Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions.
In *Advances in Neural Information Processing Systems*, pages 4765–4774, 2017.

$$v_{\mathbf{x}}(A) = \mathbb{E}_{p_{\bar{A}|A}} [f(\mathbf{x}_A, \mathbf{X}_{\bar{A}})]$$

This is a conditional expectation depending on $p_{\bar{A}|A}$.

Evaluating the influence of features



Given dataset, the model f , and the observation $\mathbf{x}$,
compute the influence $\phi_i(v_{\mathbf{x}})$ of feature $i \in N$ accessing $v_{\mathbf{x}}$ online.

Shapley value (classic)

$$\varphi_i^S(v) = \sum_{A \subseteq N \setminus i} \frac{1}{n} \binom{n-1}{|A|}^{-1} (v(A \cup i) - v(A))$$

Banzhaf value (rarely explored!)

$$\varphi_i^B(v) = \sum_{A \subseteq N \setminus i} \frac{1}{2^{n-1}} (v(A \cup i) - v(A))$$

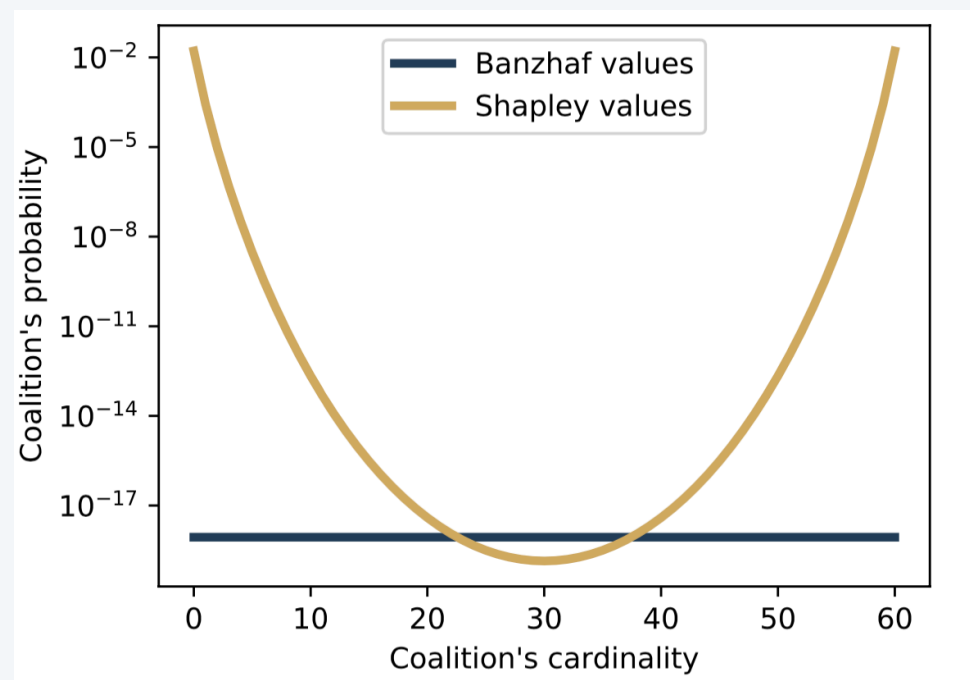
P. Somol, J. Grim, and P. Pudil. Fast dependency-aware feature selection in very-high-dimensional pattern recognition. *IEEE International Conference on Systems, Man and Cybernetics*. 2011, pp. 502–509

Shapley and Banzhaf as probabilistic values



$$p_i(A) = \frac{1}{n} \binom{n-1}{|A|}^{-1}$$

$$p_i(A) = \frac{1}{2^{n-1}}$$

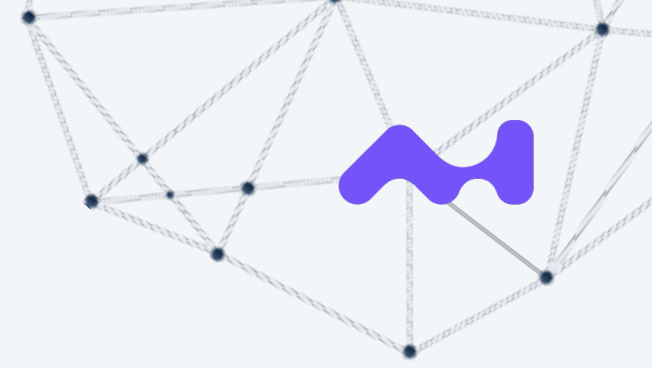




Shapley vs Banhzaf: Experiments

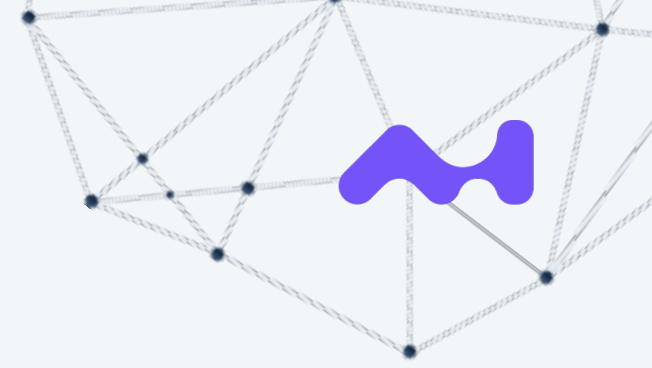
Cooperative game theory for machine learning tasks, Master thesis of J. Pecka

The setup: 3 models and 3 datasets



- For each dataset/model pair, we randomly selected 200 instances
- This defined 200 coalitional games and thus 200 Shapley/Banzhaf vectors
- We calculated Spearman's ρ and Kendall's τ to capture their *correlation*
- We observed the distribution of ρ s and τ s for each dataset/model pair

3 datasets from the UCI ML Repository



1. Titanic $d = 891$, $n = 8$: *Who survives?*
2. Breast cancer $d = 569$, $n = 30$: *Malignant/Benign*
3. MNIST $d = 1797$, $n = 8 \times 8$: *Which digit is that?*

Models

Feedforward NN

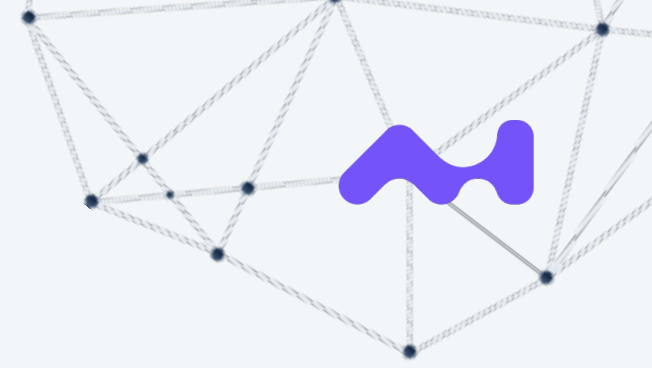
- A multilayered perceptron with 4 hidden layers (50, 50, 50, 30)
- Adaptive learning rate and ReLU activation functions

Gradient Boosted Decision Tree

- 100 gradient boosted decision trees
- Each at most 5 layers deep
- The learning rate is set to 0.1

SVM

- Gaussian kernel
- Regularization parameter 1 and $\gamma = 1/d$



The correlations between Shapley and Banzhaf



Table 3.2: Percentiles of correlation coefficients for the Titanic dataset.

	NN		GBDT		SVM	
	ρ	τ	ρ	τ	ρ	τ
mean	0.96	0.91	0.98	0.95	0.96	0.91
min	0.61	0.50	0.76	0.57	0.67	0.50
5%	0.86	0.78	0.90	0.79	0.78	0.64
10%	0.90	0.79	0.95	0.86	0.83	0.71
25%	0.95	0.86	0.98	0.93	0.95	0.86
50%	0.98	0.93	1.00	1.00	0.98	0.93
75%	1.00	1.00	1.00	1.00	1.00	1.00
90%	1.00	1.00	1.00	1.00	1.00	1.00
95%	1.00	1.00	1.00	1.00	1.00	1.00
max	1.00	1.00	1.00	1.00	1.00	1.00

Table 3.4: Percentiles of correlation coefficients for the digits dataset.

	NN		GBDT		SVM	
	ρ	τ	ρ	τ	ρ	τ
mean	0.83	0.71	0.84	0.74	0.80	0.69
min	0.67	0.54	0.67	0.56	0.57	0.44
5%	0.74	0.60	0.75	0.63	0.68	0.57
10%	0.75	0.61	0.76	0.65	0.70	0.59
25%	0.79	0.65	0.80	0.69	0.75	0.64
50%	0.85	0.72	0.85	0.75	0.81	0.69
75%	0.89	0.77	0.89	0.78	0.85	0.75
90%	0.91	0.79	0.92	0.82	0.90	0.79
95%	0.92	0.82	0.93	0.84	0.91	0.81
max	0.95	0.85	0.97	0.91	0.94	0.86

The issues of explainability



1. *The Banzhaf value is not usually considered in literature for explaining the models' behavior. Why?*
2. And why not any other *semivalue*?
3. Or a *quasivalue*?
4. How to assess the similarity of rankings for the features?